THE FIVE PHYSICAL COLUMNS OF EMBODIED INTELLIGENCE

The Fundamental Physical Bounds Governing Every Closed-Loop Cyber-Physical System

COLUMN 1

TIME & FRESHNESS

τ_world Latency Bounds

- Sense-to-actuation delay Δt
- P99.9 tail latency freshness
- Stale observations cause drift
- Hard deadlines: $\Delta t \leq \tau$_world

COLUMN 2

INERTIA & MOMENTUM

$p = mv$ Kinetic Energy

- Mass cannot stop instantaneously
- Kinetic braking distance d_stop
- Centrifugal & Coriolis forces
- Work envelope geofencing

COLUMN 3

ACTUATION DYNAMICS

τ_motor & Friction Limits

- BLDC torque-speed envelopes
- Gearbox backlash & jerk bounds
- Coulomb/viscous joint friction
- Smooth C^2 acceleration profiles

COLUMN 4

ENERGY & THERMODYNAMICS

$P = IV$ & Joule Heat

- Battery voltage sag under load
- Junction heating $T_j = T_a + P \cdot \theta_{JA}$
- DVFS thermal throttling cascades
- Regenerative braking limits

COLUMN 5

SILICON & COMPUTE

SRAM, AXI & DMA Memory

- Zero dynamic malloc in RT loops
- UMA crossbar bus contention
- Hardware watchdog supervisors
- Deterministic FreeRTOS priority

FIRST-PRINCIPLES PHYSICAL GUARANTEE

Software models propose intent; the Five Physical Columns enforce the boundary of physical possibility.